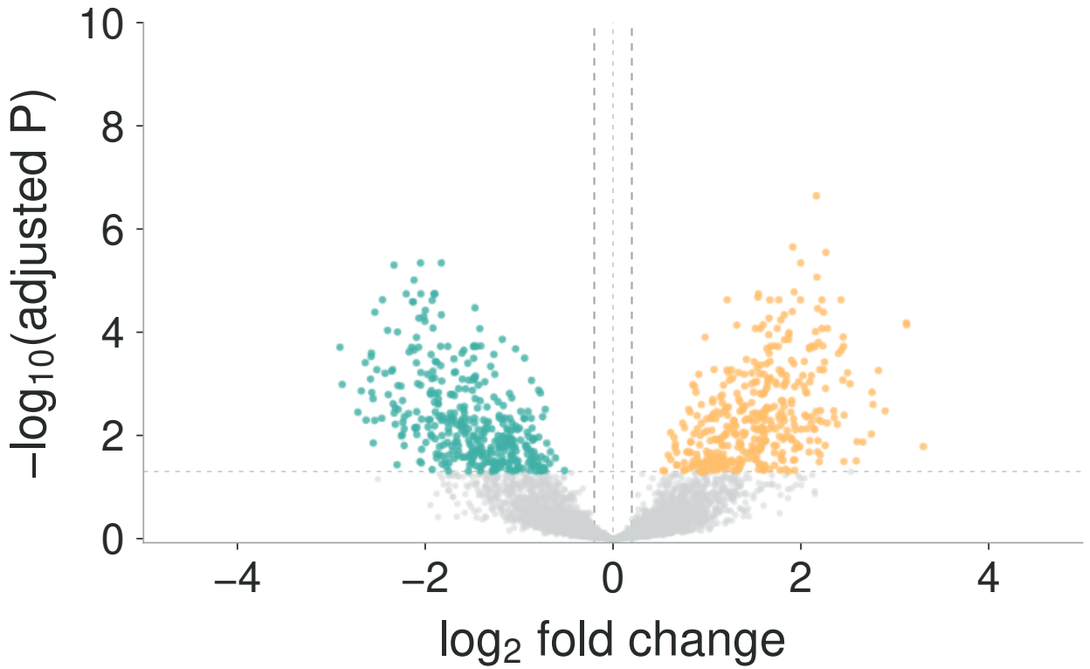
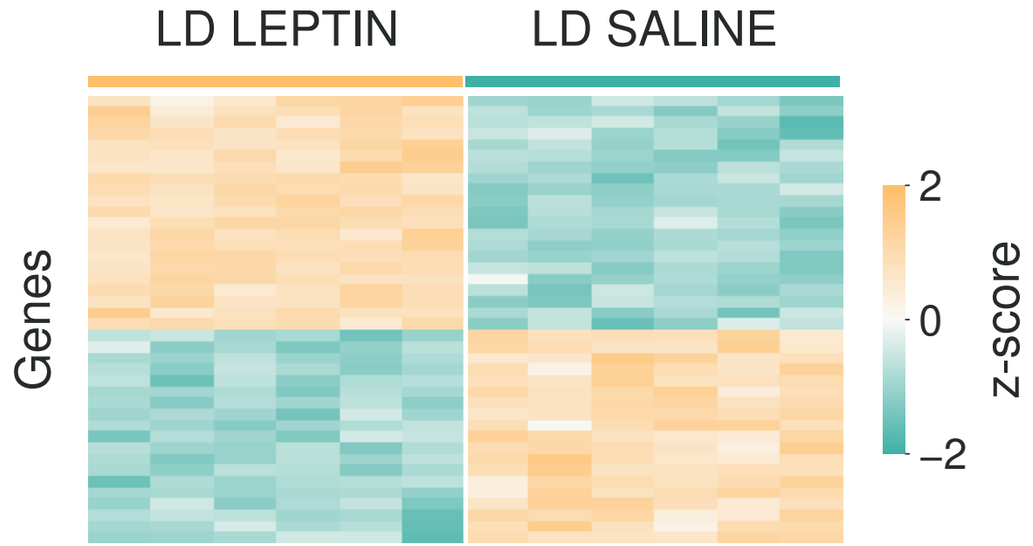
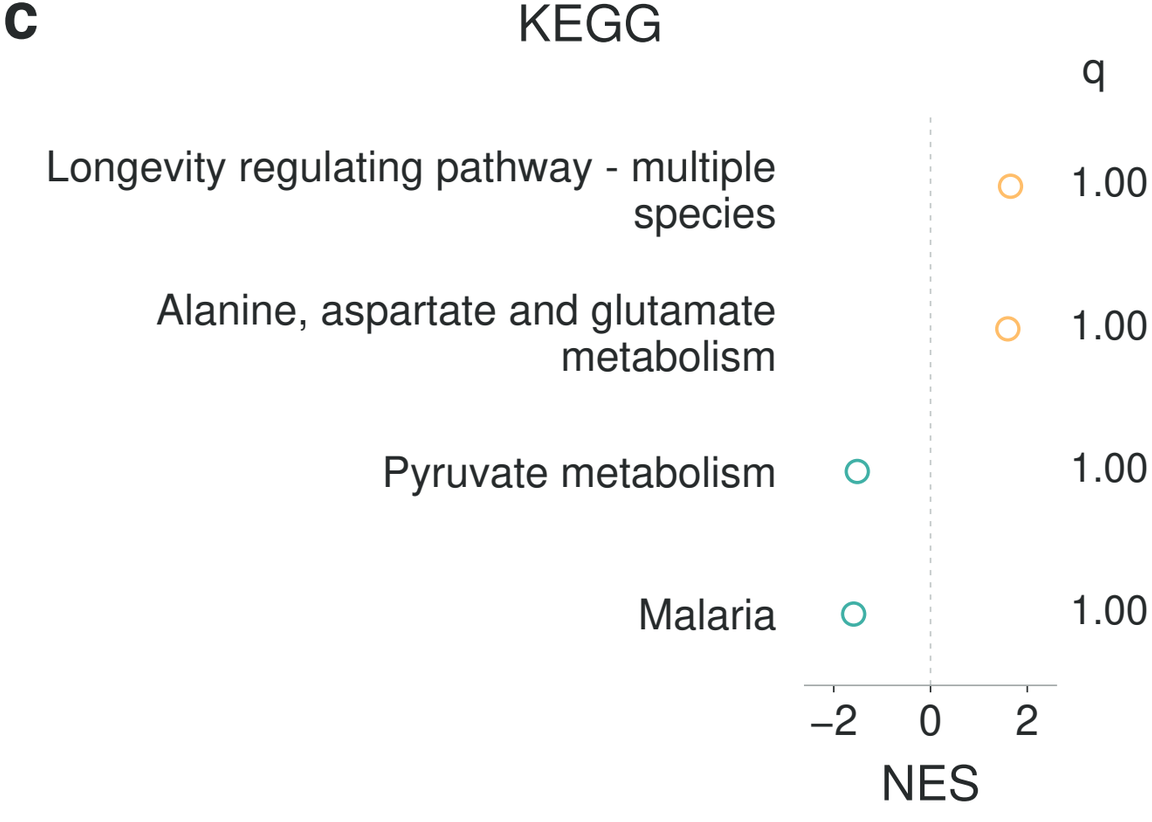
**a**


LD LEPTIN vs LD SALINE
10
8
6
4
2
0
−log10(adjusted P)
−4
−2
0
2
4
log2 fold change

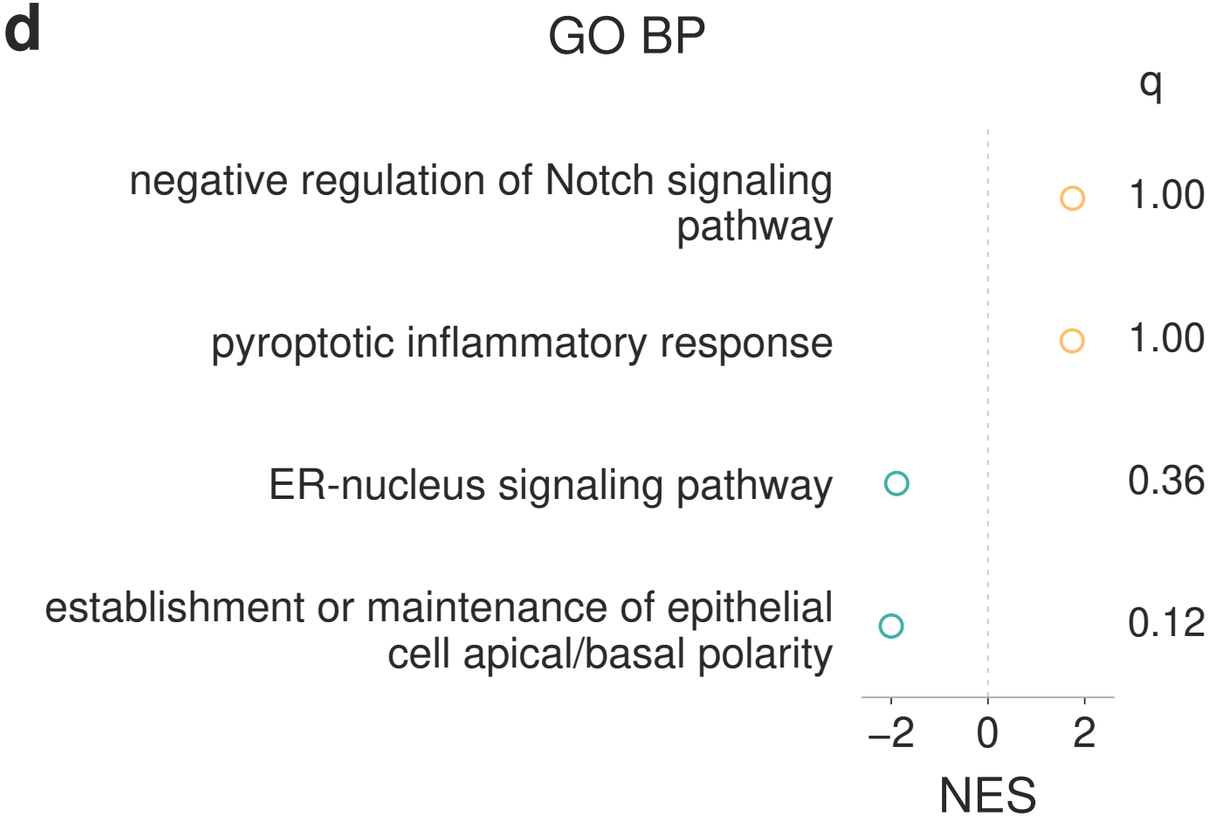
**b**


LD LEPTIN
LD SALINE
Genes
2
0
−2
z-score

**c**


KEGG
q
Longevity regulating pathway - multiple species
1.00
Alanine, aspartate and glutamate metabolism
1.00
Pyruvate metabolism
1.00
Malaria
1.00
−2
0
2
NES

**d**


GO BP
q
negative regulation of Notch signaling pathway
1.00
pyroptotic inflammatory response
1.00
ER-nucleus signaling pathway
0.36
establishment or maintenance of epithelial cell apical/basal polarity
0.12
−2
0
2
NES